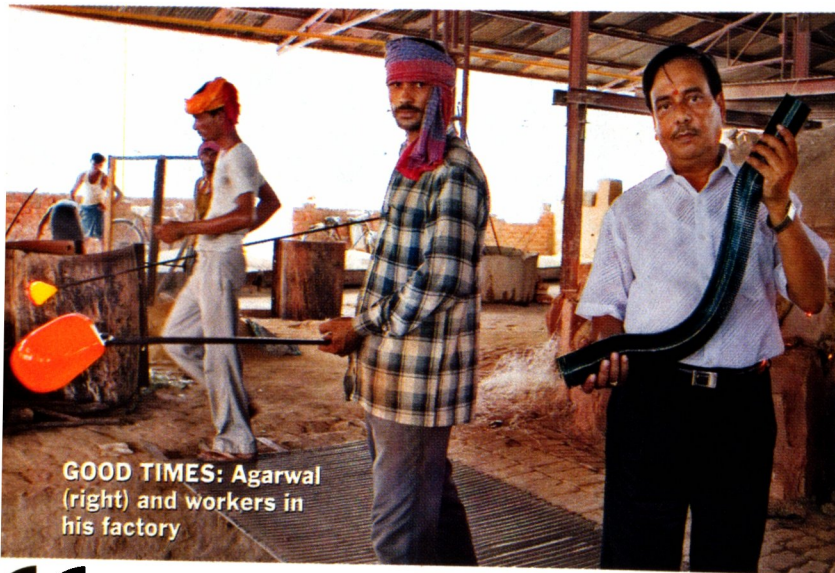


support of the Swiss Agency for Development and Cooperation (SDC), was developing clean, energy-efficient technologies based on natural gas for pot furnaces. Their model cut energy consumption by nearly 30 per cent compared to the other retro-fitted gas-fired pot furnaces. The challenge, however, was getting the factory owners involved. The owners would come for the meetings, but weren't interested, says TERI spokesperson Rajiv Chibber. The reason, says Khan, was the cost. The old coal furnace cost owners between Rs 1-2 lakh but the TERI furnace came for Rs 30-40 lakh. "If the technology flopped, my factory would shut down. But the reward was greater if it worked," says Khan.



GOOD TIMES: Agarwal (right) and workers in his factory

“Earlier, if you walked into a factory in a white shirt, you’d come out all blackened. Now, the air is cleaner and my workers are healthier.”

LALITESH K. AGARWAL, Factory owner

Persuasion finally worked. Khan was the first one to take the plunge. TERI and SDC agreed to bear hardware and construction costs as well as production expenses during the trial run while Khan agreed to bear production costs once the furnace was stabilised and to buy it back, at a concession, after a successful demonstration.

Now, standing near the TERI furnace,

Khan barks out instructions to his workers. It's close to 11 a.m. and there's a cool breeze outside. But inside, next to the furnace, it feels like a hot afternoon in the Sahara desert. Khan wipes the sweat off his face, as he monitors his workers even as he reminisces the trials of a new technology at work. It took nearly two years for the problems to be ironed out. But the one challenge that gave him many sleepless nights was the colour of the bangles. His factory is known for the deepest red colour called Lal Anju—a recipe that was given by a stranger to his grandfather Bhooray Khan. The new furnace was giving problems on this front. Once problems of clarity had been worked out, Khan realised there was inconsistency in the colour. "Red is the most sensitive hue in glassmaking," he says. In a report released this year, TERI's Girish Sethi, who headed the project, notes, "The entire Firozabad glass industry was looking eagerly at us. The technology that we were demonstrating had not been tried out anywhere else in the world." But Khan didn't give up. After experimenting on various chemical combinations, he finally managed to get the colour right.

And while Khan may have looked towards the future of a cleaner fuel and lower costs, the others were still apprehensive. It was in 2004 that Lalitesh Kumar Agarwal, owner of Bapu Glass Industries, spoke to Khan about shrinking profits. Khan gave a demonstration of the TERI furnace to Agarwal, who reluctantly got one installed. It was a noticeable change. "Earlier, if you walked into the factory in a white shirt, you'd come out all blackened. Now, the air is cleaner, my workers are healthier," says Agarwal who went on to install three more TERI furnaces. Since 2000, 40 out of the around 80 glass bangle factories in Firozabad have adopted the TERI furnace and there are more in installation stages. Some units have even drawn upon the model to develop their own energy efficient, heat recovery devices. "The Firozabad glass bangle factory initiative has become a flagship project," says Chhibber, for the ongoing TERI project Competence Network for Small and Micro Learning Enterprises (COSMILE).

And while the Firozabad experience has become an important case study for TERI, Khan is busy reaping its rewards. "Do you know one bangle passes through 45 hands," he says. Make it 45 hands plus one innovative piece of technology that brought life back into the glass bangle industry. ■